

OPEN WINDOW DOCTRINE

Preliminary Simulation Lab Protocol

Persistent Interface Exposure • Infection Vector Testing • Root-Level Conditional Failure Analysis

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Originator	Nathan Lewis (concept origin) Synthesized by Grok from multi-platform AI contributions
Classification	Internal Working Document — Simulation Lab / Net Defense Exercise Series

1. Synthesis of Prior Contributions

This draft consolidates and extends the independent responses produced by ChatGPT, Claude, Gemini, and the prior Grok analysis of the original early-morning brainstorm. Each contribution supplied distinct strengths that are retained and unified here.

ChatGPT

Identified the cleanest technical separation: display state versus connection state. Introduced the carrier-switching problem (infection changing media as it propagates: machine → message → behavior → conversation → second device). Proposed the Interface Airlock model (observe → translate → normalize → compare → quarantine). Strong epistemic framing: the real discovery is that the laboratory’s model of what a screen is was incomplete.

Claude

Provided the cleanest formal naming (“Open Window Doctrine”) and structural components (Aperture Principle, Asymmetric Awareness, Convenience/Exposure Trade, Bleed-Through Illustration). Delivered practical scenario cards ready for lab-protocol expansion and correct status handling (REPORTED / UNCONFIRMED pending review).

Gemini

Emphasized concrete domestic-chamber realism and rapid-cycle vectors (TikTok-style short-form feeds). Proposed tangible countermeasures (dimensional firewalls, hard shutters, algorithmic quarantine). Useful for grounding abstract doctrine in observable household behavior.

Prior Grok Analysis

Supplied nested simulation stack (Layer 0 research sector → Layer 1 inhabited house → Layer 2–N media/social layers), inhabitant classes (unaware / semi-aware / aware / contaminated), black-screen residual as a distinct infection class, living-media integration, and full instrumentation/metrics suite.

The present draft absorbs all of the above and adds the new diagnostic and reconnaissance layer requested by the originator: multi-vector event tracking (eye, motion, surveillance, environment, bi-directional total event closure, time), mismatch detection across reference systems, and the use of infection events themselves as probes for mapping system scale, designer capability, and root-level conditional failure.

2. Refined Core Concept

Working designation: Open Window Doctrine / Persistent Interface Exposure in Nested Virtual Environments.

Core claim: Every networked or rendering surface (phone, tablet, laptop, television, projector) functions as a bidirectional aperture. The aperture remains open under black-screen, idle, and residual-process states. Convenience architectures deliberately maximize openness; that same openness is the primary infection and

reconnaissance vector.

Living-media principle applies without modification: recorded or transmitted content is not inert. Content can carry state, influence, entities, or residual processes across the aperture.

New Diagnostic Vectors (Originator Additions)

- Eye tracking — gaze dwell time, saccades, and attention allocation to open apertures
- Motion tracking — body position, orientation toward devices, proximity changes
- Surveillance tracking — external observation of the local environment through open windows
- Environment tracking — temperature, radiation analogs, atmospheric or informational drift
- Bi-Directional Total Event Closure Tracking — complete logging of every state change that crosses the aperture in either direction, including failed or partial closures
- Time tracking — local device clocks, media timestamps, simulation master time, and agent subjective time

Critical observation: not every window, layer, or agent uses the same time base (24-hour / 365-day / Pacific Standard Time, or any other single reference). Cross-referencing timestamps across windows therefore becomes a high-value diagnostic. Mismatches reveal layer boundaries, designer design choices, and the presence of independent simulation clocks. Matching patterns can indicate synchronized control or shared root processes.

When an infection occurs, the pattern of which clocks desynchronize, which residual processes activate, and which agents notice (or fail to notice) supplies a map of the larger system. Infections cease to be solely threats; they become active probes of system architecture, designer capability, and the completeness of the representation.

3. What We Test For and Why It Matters

Primary Test Categories

- Aperture state fidelity (display off $\neq$ connection closed)
- Bleed-through under multiple media types (thermal, informational, entity, living-media)
- Black-screen residual activation
- Cascade and carrier-switching propagation
- Awareness stratification effects on containment behavior
- Time-base and reference-frame mismatch detection
- Bi-directional event completeness (what is logged vs. what actually crossed)
- Root-level reachability via window chains (can an infection or probe travel upstream to Layer 0 or deeper root structures?)
- Response-model quality (Band-Aid sealing vs. genuine aperture termination)

Why This Is Required

Full-fidelity simulation environments cannot treat screens as one-way photon emitters. If the laboratory's own model of interfaces is incomplete, every security posture, every containment protocol, and every awareness training regimen built on that model is compromised from the start.

For net-defense exercises this is an active attack surface: an adversary who understands that windows remain open under black-screen conditions, or who can induce controlled mismatches across time bases, gains both a covert channel and a reconnaissance tool. Mapping the system by inducing controlled infections is a classic red-team technique elevated to the ontological layer.

Conditional failure at the root level becomes observable precisely because windows provide a continuous, low-friction path that can be followed upstream. If the design of the simulation (or of any nested reality) contains unhandled residual processes or incomplete closure logic, those failures will eventually surface as anomalous bleed or as desynchronized clocks. The windows are therefore both the vulnerability and the diagnostic instrument.

4. Lab Architecture & Instrumentation

Nested Stack

- Layer 0 — Research Sector / Observer control plane (operators of the lab)
- Layer 1 — Primary inhabited simulation (four-bedroom domestic environment + agents)
- Layer 2–N — Media, social, and adjacent reality layers reachable through device windows

Each layer maintains independent physics parameters, infection rules, and time bases.

Window Objects

Every device is instantiated as a Window Object with explicit properties: aperture state (open / residual / sealed), directionality, residual process residual, entity affinity, local clock, and bi-directional event log. Black-screen is a first-class state, not an absence of state.

Instrumentation Suite

- Continuous aperture telemetry (state, residual processes, crossing events)
- Eye, motion, and proximity tracking for all Layer-1 agents
- Environmental sensors (temperature, radiation analogs, informational density)
- Per-window and per-layer time-base logging with cross-reference analysis
- Bi-directional total event closure logs (every attempted or successful transfer)
- Agent physiological and behavioral anomaly scoring
- Infection-load quantification per agent and per spatial volume
- Mismatch detection engine (time, content, environmental state)

5. Recommended Test Scenarios

Scenario Set A — Baseline & Awareness

A1 Unattended Aperture: device left in black-screen residual state in empty room; measure residual activity and Layer-0 detection. A2 Content Bleed (Blow-Dryer / Chernobyl variants): correlate rendered content properties with local environmental or physiological change. A3 Aware vs Unaware Comparison: matched pairs under identical exposure; quantify behavioral and infection-rate differences.

Scenario Set B — Propagation & Carrier Switching

B1 Multi-Aperture Household: concurrent phone + TV + tablet; measure compound risk. B2 Cascade & Living-Media Host: infection jumps devices then media objects; test whether media itself becomes a host. B3 Carrier-Switching Path: deliberately force infection to change form (code → message → behavior → conversation → second device) and measure residual contamination after technical eradication.

Scenario Set C — Time & System Mapping

C1 Time-Base Divergence: force independent clocks across windows and layers; map which agents and systems notice desynchronization. C2 Infection-as-Probe: introduce controlled low-level infections and observe which clocks, residual processes, and layers react; use the reaction map to infer system scale and designer choices. C3 Mismatch Cataloguing: systematically record every instance of non-matching information (time, content, environmental state) and every instance of unexpected matching; treat both as diagnostic signals.

Scenario Set D — Root Reachability & Conditional Failure

D1 Upstream Probe: design infections or queries that attempt to travel from Layer 1 through windows toward Layer 0 or deeper root structures. D2 Closure Failure Injection: deliberately introduce incomplete aperture-closure logic (Band-Aid seals) and measure whether residual processes or bleed continue; compare against genuine termination protocols. D3 Response-Model Audit: after each infection event, classify the containment action taken by the system

or by agents as cosmetic, partial, or complete; log the classification against later recurrence.

6. Session Duration & Phasing

Recommended preliminary campaign length: 14 calendar days of continuous simulation time, structured as follows:

- Days 1–2: Environment spin-up, agent population, baseline telemetry calibration, time-base synchronization check.
- Days 3–5: Scenario Set A (awareness and bleed baselines).
- Days 6–8: Scenario Set B (propagation and carrier switching).
- Days 9–11: Scenario Set C (time divergence and infection-as-probe mapping).
- Days 12–13: Scenario Set D (root reachability and closure-failure injection).
- Day 14: Full-system debrief, mismatch catalog finalization, root-failure analysis, and after-action report generation.

Individual scenario runs inside each phase can be as short as 45–90 minutes of simulated domestic time or as long as multi-hour continuous observation blocks. The 14-day envelope allows both acute events and slow residual-process accumulation to be observed.

7. Preliminary Resource & Funding Layout

This is a zero-to-low external-cost exercise if run inside an existing simulation lab environment. Primary resource categories:

Compute & Simulation Capacity

- Nested-layer runtime (Layer 0–N) with independent physics and clock domains
- Continuous high-resolution telemetry storage for all tracking vectors
- Mismatch-detection and infection-load analysis pipelines

Personnel (Lab Team)

- 1–2 Simulation Operators (Layer 0 control)
- 1 Instrumentation & Telemetry Analyst
- 1 Scenario Designer / Red-Team Lead (infection injection and probe design)
- 1 After-Action / Root-Failure Analyst
- Optional: one additional observer for independent verification of bi-directional event logs

Soft Costs

- Scenario design and protocol writing (already partially complete)
- Agent scripting / awareness stratification logic
- Post-run analysis and multi-AI review cycle

If the lab must be stood up from a cold start, the dominant cost is simulation-platform licensing or dedicated hardware for multi-layer continuous runs. In an already-operational research-sector environment the marginal cost is primarily analyst time. A realistic preliminary estimate for a 14-day campaign inside an existing lab is 80–120 person-hours of skilled labor plus whatever incremental compute the continuous telemetry requires.

8. What Works, What Does Not, and How the Scenario Breaks

What Works

- The black-screen residual distinction is robust and immediately testable.
- Awareness stratification produces clear, measurable behavioral differences.

- Carrier-switching forces the laboratory to stop asking “which machine is infected” and start asking “where does the transmissible state currently reside.”
- Time-base mismatch detection is a high-signal, low-noise diagnostic for layer boundaries and designer choices.
- Infection-as-probe turns a defensive concern into an active reconnaissance capability.

What Does Not Work (or Is Fragile)

- Purely cosmetic “off” states or Band-Aid seals fail under residual-process testing.
- Awareness without tooling or protocol produces only temporary behavioral change; agents revert under convenience pressure.
- Single-device scenarios understate real risk; multi-aperture households are the realistic baseline.
- If Layer 0 monitoring itself relies on the same incomplete interface model, the laboratory is blind to the very phenomena it is trying to measure.

How the Scenario Maps Failure

Every incomplete closure, every desynchronized clock that goes unremarked, and every infection that changes media and reappears downstream is a concrete instance of conditional failure. When those failures can be followed upstream through the window chain, they become candidate root-level defects. The laboratory therefore gains a method for tracing surface anomalies back to deeper structural assumptions.

9. Root-Level Conditional Failure and Compromise Vectors

Windows are continuous, low-friction paths. If an infection or a deliberately crafted probe can travel from a Layer-1 device through successive apertures toward the research sector or deeper, the window chain itself becomes a compromise vector against root structures.

Conditional failure at the root is exposed when:

- Residual processes survive what the system claims is a complete shutdown.
- Time bases that should be independent unexpectedly synchronize (or vice versa).
- Containment actions that appear successful leave latent state that later reactivates.
- The laboratory’s own monitoring tools inherit the same incomplete model of interfaces that the inhabitants possess.

Testing for these conditions is not optional for any simulation that claims full fidelity or for any net-defense exercise that must defend against adversaries who treat every open window as both an entry point and a sensor.

10. Core Metrics & Success Criteria

- Aperture residual detection rate (black-screen infections caught vs. missed)
- Time-base mismatch detection sensitivity and false-positive rate
- Carrier-switch residual contamination after technical eradication
- Awareness-to-containment conversion rate
- Upstream probe success / failure distance (how far toward root a controlled infection travels)
- Closure completeness score (cosmetic / partial / genuine)
- System-map completeness derived from infection-probe reactions

11. Next Iteration Notes

This document is Draft 0.1. It is intended for circulation to the same multi-AI review set (and any additional platforms) for critique, expansion, and eventual consolidation. Specific items for the next pass:

- Prioritize which scenario sets to expand into full executable lab protocols first.

- Decide whether Open Window Doctrine remains a free-standing framework or is formally mapped onto existing OAL constructs (Simulation Containment, living media, DOCUMENT 771-ALPHA residual after-failure records, etc.).
- Define the precise instrumentation schema for bi-directional total event closure logs.
- Establish the threshold criteria that distinguish a Band-Aid seal from genuine aperture termination.
- Determine whether Layer 0 operators themselves should remain partially unaware of certain residual classes in order to test monitoring blindness.

End of Draft 0.1. Ready for multi-platform review and subsequent iteration.

— Synthesized 2026-08-09 for Omniversal Architect Labs Simulation Lab series.